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Dynamic Analysis of a Multi-Shaft Helical Gear Transmission by Finite Elements: Model and Experiment

A dynamic model of a multi-shaft helical gear reduction unit formed by N flexible shafts is proposed in this study. The model consists of a finite element model of shaft structures combined with a three-dimensional discrete model of helical gear pairs. Bearing and housing flexibilities are included in the model as well. Eigenvalue solution and the Modal Summation Technique are used to predict the free and forced vibrations of the system. Results of experimental study on a helical gear-shaft-bearing system are also presented for validation of the model. It is demonstrated that the predictions match well with the experimental data in terms of excited modes and the forced response given in the form of the dynamic transmission error. Forced vibrations of an example system formed by three shafts are also studied to demonstrate the influence of some of the key system parameters. [DOI: 10.1115/1.1760561]

1 Introduction

Dynamic analysis of gear systems is essential in describing noise and durability characteristics of any geared power transmission system. Forces acting on gears are amplified under dynamic conditions resulting in larger dynamic loads and stresses that potentially reduce the fatigue life of the gears. In addition, high-frequency alternating forces due to of the dynamic behavior are transmitted to the radiating surfaces, forming the excitation for structure-borne gear whine. Therefore, a dynamic model for prediction of vibration amplitudes and dynamic forces is necessary to assess a given gear system for noise and durability, and to identify and implement potential solutions to either problem.

Many different types of gear systems can be found in automotive, aerospace and industrial applications while the most common ones are counter-shaft (parallel-axis) gear sets, cross-axis gear sets and planetary gear sets. Counter-shaft gear sets are the main focus of this study. Each counter-shaft system is formed by a number of flexible parallel shafts supported on flexible bearings at certain locations. The shafts are connected to each other through spur or helical gear pairs. One common configuration of such systems is shown in Fig. 1 schematically. This system finds its applications in many products including passenger car manual transmissions, automatic transmission final drive units, continuously variable transmission final drive units and industrial gear reduction gearboxes.

Dynamics of gear systems has attracted the attention of many researchers over the years. Most of the earlier dynamic models focused on a single spur gear pair. Papers by Ozguven and Houser [1] and Blankenship and Singh [2] offer a comprehensive review of these models. One group of gear pair studies [3–7] is of primary interest here as it forms the basis for the modeling of systems like the one illustrated in Fig. 1. These models considered a single spur gear pair supported by flexible shafts and bearings. The shaft flexibilities were represented by formulations of varying sophistications, from simple equivalent lumped springs to finite element representations. Free and forced vibration results from these studies revealed a strong coupling between transverse and torsional vibrations, suggesting that influence of shaft and bearing

flexibilities are indeed important. Another group of studies [8–10] investigated the dynamic behavior of helical gear pairs. These studies focused on three-dimensional vibrations caused by the helical gear interface where a 12-degree-of-freedom model of the helical gear pair model was supported by a simplified support flexibility model to demonstrate the complex coupling amongst the transverse, torsional, axial and rotational motions of gears.

There are a number of published papers that specifically studied the system shown in Fig. 1 for a 3-shaft configuration. Early examples of these models were mostly torsional like the model by Iida [11]. More recent works on modeling of this system [12–15] included the shaft and bearing flexibilities to a certain extent. For instance, the model by Lim and Li [13] proposed a $12n$ -degree-of-freedom model of an n gear ($n - 1$ shafts) system with lumped shaft flexibilities. Raclot and Vexé [15] combined a similar model of helical gear pairs with a Ritz-based formulation of the shaft flexibilities. Similarly, Vinayak et al. [14] used a helical gear pair model similar to the one developed by Blankenship and Singh [9] in conjunction with the a lumped shaft-bearing model. Kahraman [16] developed a helical gear model of the system in an idler configuration to study the differences between the dynamic behavior an idler system and a split-torque system. Since these models were concerned mostly with the gear mesh dynamics rather than the force transmissibility or design of shafts and bearings, detailed models of the shafts and bearings were not required.

The first objective of this study is to develop a dynamic model of a multiple counter-shaft helical gear system. As the focus is not only on the gears but also on the shafts and bearings, an equal amount of attention will be given to an accurate representation of shafts and bearings as well. Therefore, the model must have the capability of including the most relevant design parameters associated with the gears, shafts and bearings in order to quantify the direct influence of each of these parameters on the overall dynamic behavior. The second objective is to validate the dynamic model predictions. For this purpose, results of a detailed experimental study will be presented and compared to the predicted forced response of the same experimental gear set. At the end, limited parametric studies will be performed to describe the effect of some of the key system parameters on the dynamic behavior. The intent of this model is to aid designers at any stage of the development of the system to guide them to the most favorable configuration for the most desirable dynamic behavior. Therefore,

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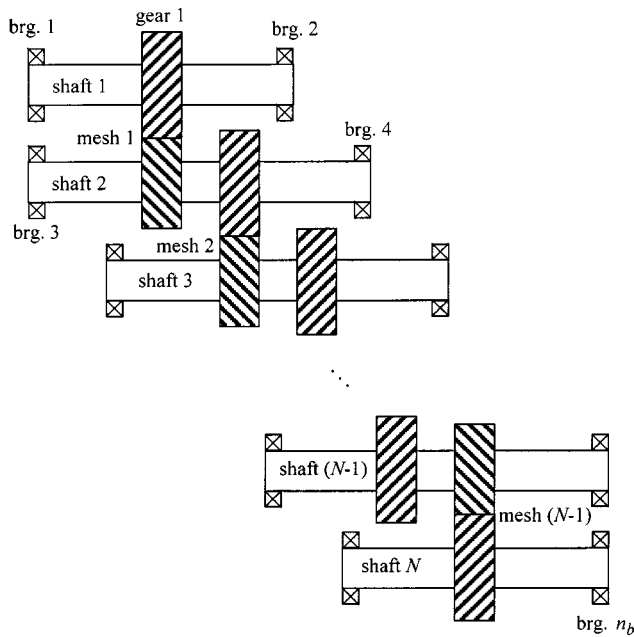


Fig. 1 A counter-shaft system having N shafts, $N-1$ gear meshes and n_b rolling element bearings

as a computer aided design tool, the model must be as general as possible to analyze different systems and configurations.

2 Dynamic Model

2.1 Model Capabilities and Assumptions. A coupled three-dimensional (transverse-rotational-axial) model of the systems shown in Fig. 1 will be proposed in this study. The formulation will be presented for the most general case of N shafts, connected to each other through $(N-1)$ gear meshes formed by $2(N-1)$ gears, while most automotive and industrial applications contain up to four shafts ($N \leq 4$). The model must be general to include: (i) vibrations of helical gears and shafts in all possible directions (transverse, torsional, axial and rotational), (ii) any number of shafts of arbitrary geometries including variable diameters, solid or hollow cross-sections, (iii) any couplings, flywheels, sprockets and other rigid inertias that are mounted on the shafts including the inertias of the drive unit and the load, and (iv) any number of rolling element bearings of any type supporting the shafts at given locations.

The proposed model is based on an existing geared rotor dynamics model developed by one of the authors [5]. This previous model considered a coupled spur gear-shaft-bearing dynamics of a two-shaft, single-gear pair system. A lumped mass model of the spur gear pair was combined with a finite element model of the shafts to predict the natural modes and the forced response due to static transmission error excitation. As an extension of this earlier model, the model presented here adds axial and rotational motions of the shafts and also helical gears for a system formed by N shafts as shown in Fig. 1. The model will be designed such that any system of the same basic layout can be analyzed to allow parametric design sensitivity studies.

The proposed model assumes that the dynamic behavior is linear. Any nonlinear effects originating from tooth separations due to gear backlash and shaft separations due to bearing clearances are assumed to be negligible. This was reported to be a valid assumption especially for moderately or highly loaded high contact ratio helical gears [17] and preloaded rolling element bearings [18,19]. Also neglected are time-varying effects such as the fluctuations of the gear mesh stiffness due to alternating number of tooth pairs in contact. While gear mesh stiffness fluctuations of

spur gears are large enough to cause parametric resonances [20], they are much smaller for high contact ratio helical gears. Bearing stiffness fluctuations that are also present as the number of rolling elements in the loaded zone fluctuates [21] are also assumed to be negligible here. Validity of these two key assumptions (no tooth separations and time-invariant mesh stiffness) will be demonstrated in Section 3 through experimental data collected by using a helical gear-shaft-bearing system.

It is also assumed here that the parameters used to characterize the gear meshes, primarily the gear mesh stiffness and the motion transmission errors, remain unchanged under dynamic conditions. Ideally, one would prefer a model that includes these parameters implicitly perhaps through a deformable body contact model of the gear mesh as well so that a more realistic representation of gear mesh is possible even under dynamic conditions. Although such a deformable gear body approach was employed to a single gear pair successfully in the past [22], it was not feasible here as the system is much larger. In addition, the proposed model is intended to be a design tool to be used in different stages of gearbox design and development, preventing the use of such very demanding numerical schemes. Instead, these gear mesh parameters will be obtained under loaded quasi-static conditions using available contact mechanics models [23] and will be assumed to remain the same under dynamic conditions as well.

2.2 Model Formulation. A Finite Element (FE) formulation will be used to develop a model of shafts in Fig. 1. Here, a Timoshenko beam formulation can be employed if effects such as rotary inertia, shear deformations and gyroscopic moments are expected to be significant for the application considered [24,25] or Euler beam model might suffice if shear deformation effects are not significant [5]. The stiffness and mass matrices of each finite shaft element are determined and assembled to form stiffness $\mathbf{K}_{s_n}$ and mass $\mathbf{M}_{s_n}$ matrices of each individual shaft n ($n=1$ to N). Overall shaft stiffness and mass matrices of the system are then assembled as $\mathbf{K}_s = \text{Diag}[\mathbf{K}_{s_1} \mathbf{K}_{s_2} \cdots \mathbf{K}_{s_N}]$ and $\mathbf{M}_s = \text{Diag}[\mathbf{M}_{s_1} \mathbf{M}_{s_2} \cdots \mathbf{M}_{s_N}]$. Both are symmetric, square matrices of dimension q where $q = 6 \sum_{n=1}^N (m_n + 1)$ is the total number of degrees of freedom of the system in hand (m_n is the number of finite elements used to define shaft n).

In a typical application, each shaft is supported by at least two rolling element bearings of varying type, size and design parameters. Considering that the system has a total of n_b bearings, the overall bearing matrix can be written in terms of individual bearing stiffness matrices $\mathbf{k}_{b_i}$ ($i=1$ to n_b) as $\mathbf{K}_b = \text{Diag}[\cdots \mathbf{k}_{b_1} \cdots \mathbf{k}_{b_2} \cdots \mathbf{k}_{b_{n_b}} \cdots]$. In cases when the transmission housing holding the bearings cannot be assumed rigid, the flexibility of the housing can be included in the model through a condensed stiffness matrix $\mathbf{K}_c$ that can be obtained from a FE model of the housing for the nodes associated with the bearing locations. The condensed housing stiffness matrix not only includes additional flexibilities at the bearing locations but also brings a coupling between the different bearing locations, and can be incorporated into $\mathbf{K}_b$ using the approach of references [26,27].

A general three-dimensional dynamic model of a helical gear pair is shown in Fig. 2 [8]. The system is formed by helical gears i and j , both having rigid blanks that are connected to each other by a linear gear mesh spring k_{ij} on the plane of action (a plane tangent to both base cylinders) in the direction of the tooth normal determined by the helix angle β_{ij} . Also applied in the same direction connected in series with k_{ij} is a displacement excitation in the form of a motion transmission error $e_{ij}(t)$. The relative positions of the gears are such that the line connecting the gear centers makes an angle α_{ij} with the positive x-axis. With α_{ij} defined, the angle between the plane of action and the positive y-axis becomes ψ_{ij} as shown in Fig. 2. Since the plane of action changes direction depending on the direction of the loading, ψ_{ij} is defined as

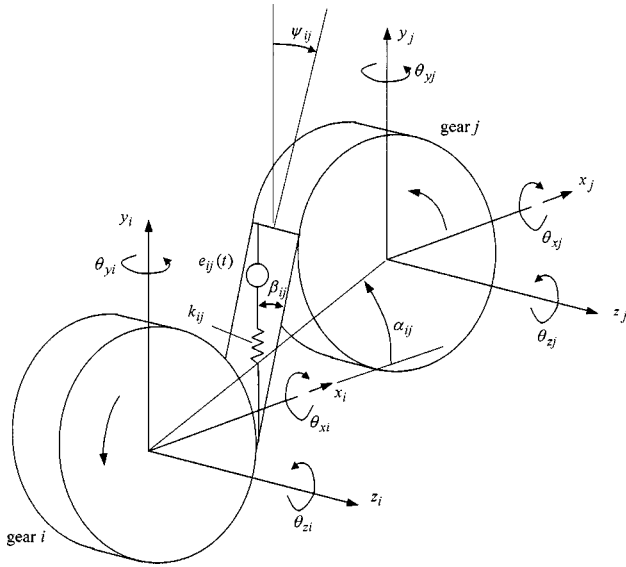


Fig. 2 Three-dimensional dynamic model of a helical gear pair

$$\psi_{ij} = \begin{cases} \phi_{ij} - \alpha_{ij}, & T_i: \text{Counterclockwise} \\ -(\alpha_{ij} + \phi_{ij}), & T_i: \text{Clockwise} \end{cases} \quad (1)$$

where ϕ_{ij} is the transverse operating pressure angle of the gear pair. The direction of the torque applied to gear i , T_i , defines the loaded flank, and hence, the direction in which the plane of action should be positioned. In addition, based on the hand of the gears, helix angle β_{ij} is defined as

$$\beta_{ij} = \begin{cases} > 0 & \text{if gear } i \text{ has left hand teeth} \\ < 0 & \text{if gear } i \text{ has right hand teeth} \end{cases} \quad (2)$$

Both gears are allowed to translate in x and y directions in the transverse plane and in the axial z direction. In addition, each gear is allowed to rotate about these three axes denoted by θ_x , θ_y and θ_z , respectively. With these six degrees of freedom for each gear, the gear pair ij has a total of 12 degrees of freedom that defines the coupling between the two shafts holding the gears. Undamped equations of motion for gear pair ij are given as:

$$m_i \ddot{y}_i + k_{ij} p_{ij}(t) \cos \beta_{ij} \cos \psi_{ij} = 0, \quad (3a)$$

$$m_i \ddot{x}_i + k_{ij} p_{ij}(t) \cos \beta_{ij} \sin \psi_{ij} = 0, \quad (3b)$$

$$m_i \ddot{z}_i - k_{ij} p_{ij}(t) \sin \beta_{ij} = 0, \quad (3c)$$

$$I_i \ddot{\theta}_{yi} + r_i k_{ij} p_{ij}(t) \sin \beta_{ij} \cos \psi_{ij} = 0, \quad (3d)$$

$$I_i \ddot{\theta}_{xi} + r_i k_{ij} p_{ij}(t) \sin \beta_{ij} \sin \psi_{ij} = 0, \quad (3e)$$

$$J_i \ddot{\theta}_{zi} + r_i k_{ij} p_{ij}(t) \cos \beta_{ij} = T_i. \quad (3f)$$

$$m_j \ddot{y}_j - k_{ij} p_{ij}(t) \cos \beta_{ij} \cos \psi_{ij} = 0, \quad (4a)$$

$$m_j \ddot{x}_j - k_{ij} p_{ij}(t) \cos \beta_{ij} \sin \psi_{ij} = 0, \quad (4b)$$

$$m_j \ddot{z}_j + k_{ij} p_{ij}(t) \sin \beta_{ij} = 0, \quad (4c)$$

$$I_j \ddot{\theta}_{yj} + r_j k_{ij} p_{ij}(t) \sin \beta_{ij} \cos \psi_{ij} = 0, \quad (4d)$$

$$I_j \ddot{\theta}_{xj} + r_j k_{ij} p_{ij}(t) \sin \beta_{ij} \sin \psi_{ij} = 0, \quad (4e)$$

$$J_j \ddot{\theta}_{zj} + r_j k_{ij} p_{ij}(t) \cos \beta_{ij} = -T_j. \quad (4f)$$

In Eqs. (3–4), the term $p_{ij}(t)$ represents the relative displacement at the gear mesh in a direction normal to contact surfaces, and it is defined as:

$$p_{ij}(t) = (x_i \sin \psi_{ij} - x_j \sin \psi_{ij} + y_i \cos \psi_{ij} - y_j \cos \psi_{ij} + r_i \theta_{zi} + r_j \theta_{zj}) \cos \beta_{ij} + (-z_i + z_j + r_i \theta_{xi} \sin \psi_{ij} + r_j \theta_{xj} \sin \psi_{ij} + r_i \theta_{yi} \cos \psi_{ij} + r_j \theta_{yj} \cos \psi_{ij}) \sin \beta_{ij} - e_{ij}(t) \quad (5)$$

With the displacement vector of gear ij defined as

$$\mathbf{x}_{ij} = [y_i \ x_i \ z_i \ \theta_{yi} \ \theta_{xi} \ \theta_{zi} \ y_j \ x_j \ z_j \ \theta_{yj} \ \theta_{xj} \ \theta_{zj}]^T, \quad (6)$$

the gear stiffness coupling matrix $\mathbf{K}_{ij}$ and the mass matrix $\mathbf{M}_{ij}$ of the gear pair ij is obtained from Eqs. (3–5). The system of N shafts shown in Fig. 2 has a total of $2(N-1)$ gears forming $(N-1)$ gear pairs. As a gear pair connects a particular node on one shaft to another node on the next shaft according to Eqs. (3–4), an overall gear stiffness and mass matrices, $\mathbf{K}_g$ and $\mathbf{M}_g$, of dimension q can be assembled.

In above equations, the individual gear mesh stiffness matrices of each mesh are used with the appropriate shaft position angles $\alpha_{(i)(i+1)}$, transverse pressure angles $\phi_{(i)(i+1)}$ and helix angles $\beta_{(i)(i+1)}$. For the N -shaft system, the angle $\psi_{(i)(i+1)}$ is given in the general form as

$$\psi_{(i)(i+1)} = \begin{cases} (-1)^{(i+3)/2} \phi_{(i)(i+1)} - \alpha_{(i)(i+1)}, & T_1: \text{Counterclockwise} \\ (-1)^{(i+1)/2} \phi_{(i)(i+1)} - \alpha_{(i)(i+1)}, & T_1: \text{Clockwise} \end{cases} \quad (7)$$

The direction of the applied input load T_1 on the first shaft again defines the angles ψ for the rest of the system since all of the planes of action of gears are changed as the input torque direction is altered.

Given the mass and stiffness matrices for shafts, bearings/housing and gears, the mass and stiffness matrices of the overall system are given as $\mathbf{M} = \mathbf{M}_s + \mathbf{M}_g$ and $\mathbf{K} = \mathbf{K}_s + \mathbf{K}_b + \mathbf{K}_g$. The alternating force vector is defined in terms of all $(N-1)$ static transmission error excitations (one at each gear mesh) as

$$\mathbf{F}(t) = \sum_{i=1,3,5}^{2N-3} \mathbf{F}_{(i)(i+1)} e_{(i)(i+1)}(t) \quad (8)$$

where $\mathbf{F}_{(i)(i+1)}$ is the extended forcing vector corresponding to gear pair $(i)(i+1)$ given by $\mathbf{F}_{(i)(i+1)} = [\cdots (\mathbf{f}_1)_{(i)(i+1)} \cdots (\mathbf{f}_2)_{(i)(i+1)} \cdots]^T$ ($i = 1, 3, 5, \dots, (2N-3)$). The static transmission error excitation functions that can be predicted by using a static gear contact mechanics model (e.g. reference [23]) can be defined in Fourier series form as

$$e_{(i)(i+1)}(t) = \sum_{r=1}^R \tilde{e}_{(i)(i+1)r} \sin[r \omega_{(i)(i+1)} t + \Gamma_{(i)(i+1)r} + r \Pi_{(i)(i+1)}], \quad i = 1, 3, 5, \dots, (2N-3) \quad (9)$$

where $\tilde{e}_{(i)(i+1)r}$ and $\Gamma_{(i)(i+1)r}$ are the r -th harmonic amplitude and phase angle and $\omega_{(i)(i+1)}$ is the gear mesh frequency of gear pair $(i)(i+1)$. While only the gear mesh components of $e_{(i)(i+1)}(t)$ are included in Eq. (9), errors associated with shaft orders can also be accounted for rather easily since the system is linear. The other phase angle $\Pi_{(i)(i+1)}$ represents the phase difference between $e_{(i)(i+1)}(t)$ and $e_{12}(t)$ where $\Pi_{12} = 0$. Considering the side view of a segment of the physical system shown in Fig. 3, one can define $\Pi_{(i)(i+1)}$ as

$$\Pi_{(i)(i+1)} = \sum_{j=1,3,5}^i \bar{\Pi}_{(j)(j+1)} \quad (10a)$$

where $\bar{\Pi}_{(j)(j+1)}$ is the phase angle between $e_{(j)(j+1)}(t)$ and the previous excitation $e_{(j-2)(j-1)}(t)$ and is given as

$$\bar{\Pi}_{(j)(j+1)} = Z_j [\bar{\alpha}_{(j)(j+1)} - \gamma_{(j-1)(j)}]. \quad (10b)$$

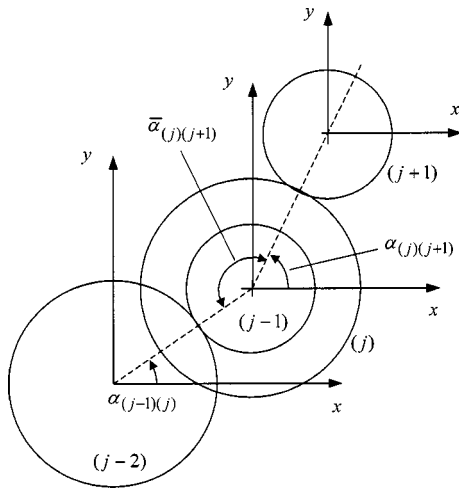


Fig. 3 Definition of shaft position angles

Here, $\gamma_{(j-1)(j)}$ is the angle between the reference teeth of gears $(j-1)$ and j that are mounted on the same shaft as illustrated in Fig. 4 and

$$\bar{\alpha}_{(j)(j+1)} = \pi - \alpha_{(j)(j+1)} + \alpha_{(j-2)(j-1)} \quad (10c)$$

2.3 Solution. Equations of motion of the overall system can be written in matrix form as

$$\mathbf{M}\ddot{\mathbf{X}}(t) + \mathbf{C}\dot{\mathbf{X}}(t) + \mathbf{K}\mathbf{X}(t) = \mathbf{F}(t) \quad (11)$$

The formulation presented up to this point did not include the derivation of the damping matrix $\mathbf{C}$. If the damping values of each component including gear meshes, bearings and shafts were known, one could obtain a damping matrix that is in the same form as $\mathbf{K}$. However, these damping values are not known in most cases even in the final stages of gear design. Therefore, for practical engineering purposes, a set of modal damping values ζ_s are used to define $\mathbf{C}$ instead of using a damping matrix that is formed by actual damping values. The eigen value problem corresponding to Eq. (11) yields the natural frequencies ω_s and the corresponding modal vectors Φ_s (mode shapes) where s is the modal index. The Sequential Jacobi Method is used here for the eigen value solutions. The forced response of the system to the forcing vector $\mathbf{F}(t)$ defined by Eqs. (8–10) is obtained by employing the Modal Summation Technique [28] as

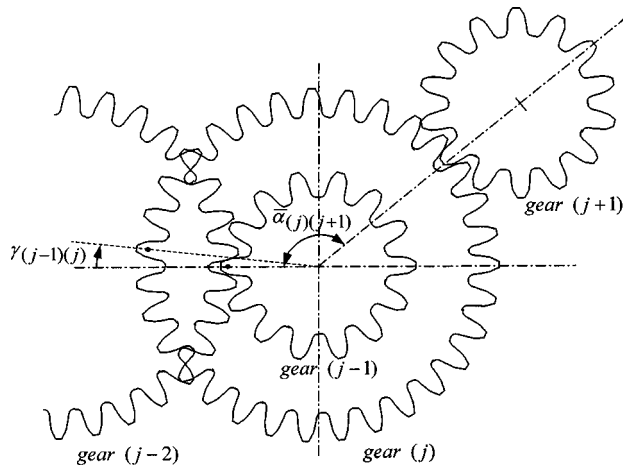


Fig. 4 Definition of phase angle $\gamma_{(j-1)(j)}$

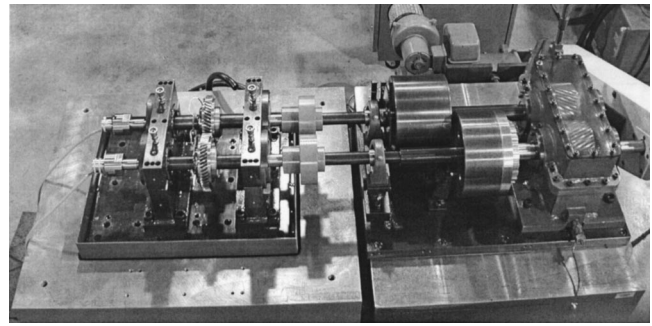


Fig. 5 Gear dynamics test machine

$$\begin{aligned} \mathbf{X}(t) = & \sum_{i=1,3,5}^{2N-3} \sum_{r=1}^R \sum_{s=1}^q \frac{\Phi_s \Phi_s^T}{(\omega_s^2 - r^2 \omega_{(i)(i+1)}^2 + 2jr \zeta_s \omega_s \omega_{(i)(i+1)})} \\ & \times \mathbf{F}_{(i)(i+1)} \{ \tilde{\mathbf{e}}_{(i)(i+1)r} \sin[r \omega_{(i)(i+1)} t + \Gamma_{(i)(i+1)r} \\ & + r \Pi_{(i)(i+1)}] \} \end{aligned} \quad (12)$$

In above equation, j is the complex imaginary number and the superscript T denotes a matrix transpose.

3 Experimental Validation

Before any parametric studies can be performed with confidence, the model must be validated through a comparison to the helical gear-shaft-bearing system experiments. Here, a set of experimental data collected using a single-mesh helical gear system will be considered, as extensive multi-mesh helical gear data are not available. Therefore, this validation effort will not address behavior unique to multi-mesh systems such as mesh phasing. However, it is still very relevant in terms of demonstrating the validity of (i) the assumptions used in the model that such helical gear systems act in a linear and time-invariant fashion (no tooth separations and constant gear mesh stiffness), (ii) the use of Euler beam model, and (iii) formulations regarding bearings and gear mesh coupling in general. At this point, a good quantitative match between the single-mesh gear experiment and model predictions of free and forced vibration behavior will provide a certain level of confidence on the multi-mesh model until a more compressive validation can be performed using multi-mesh helical gear sets data.

A four-square, power circulation type test machine shown in Fig. 5 was employed for this experimental study. The same test machine was used previously to study nonlinear behavior of spur gear pairs including jump discontinuities, parametric resonances and chaotic motions [29–31] as well as experimental investigations on the influence of certain design parameters such as tooth modifications [32] and contact ratio [33] on the dynamic behavior of spur gear pair. A detailed description of the test machine can be found in earlier these earlier papers [29–31]. It is of primary interest here to note that the test gear set shown to the left in Fig. 5 are well-isolated from the reaction gear box through massive flywheels, elastomer couplings and long, slender shafts, such that influence of the reaction gear pair on the dynamic behavior of the test gear system is negligible. A constant torque is applied to the closed loop through a split coupling.

The shafts holding the test gears have 250 mm length (distance between the centers of the bearings) and two uniform segments of 43 and 37.6 mm diameter. Each test gear is assembled on its shaft precisely to avoid any mounting error. A pair of precision spherical roller bearings supports the bearings that are housed in the bearing caps. The bearing caps and the bearings pedestals are such that there is no shaft misalignments. In addition, the bearing pedestals and the bed plate of the test machine are rigid, eliminating the need for a condensed case stiffness matrix.

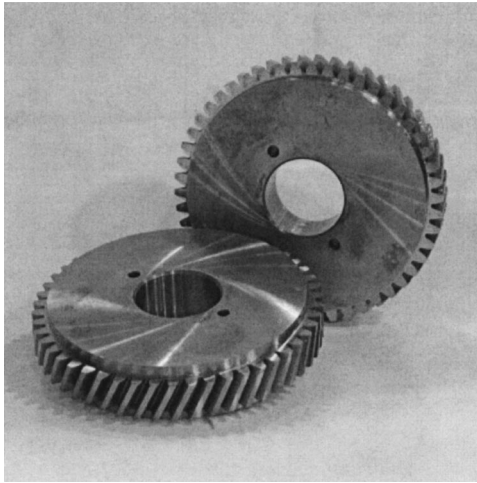


Fig. 6 A pair of helical test gears

An example helical test gear pair is shown in Fig. 6. Table 1 lists the parameters of the helical test gear pair. Three separate gear pairs of the same base designs are considered in this experimental study while the results of only two of them will be included in this paper. The difference between each gear pair is that they have different outside diameter resulting in different involute contact ratios (*ICR*) of 1.4, 1.6 and 1.775. As each gear pair has a unity face contact ratio. No tooth surface modifications are applied to the gear teeth except a very slight lead crown of about 4 μm .

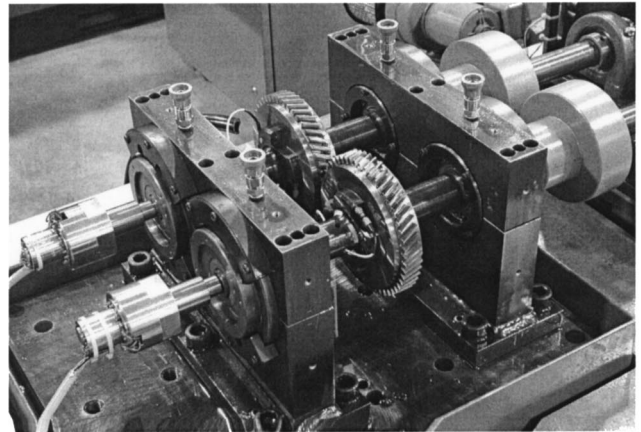
The range of the rotational speed range was 0 to 4000 rpm for each test. The dynamic transmission error of the gear pair ($DTE = r_i\theta_i + r_j\theta_j$) was used as the parameter to be used for this validation. Measurement of *DTE* within such a wide range of speed presents challenges. Encoder-based methods are capable of measuring *DTE* at lower speeds only while the accelerometer-based methods are more accurate at higher speeds due to the noise involved with double integrating the measured $\ddot{\theta}_i$ and $\ddot{\theta}_j$ at lower speeds [34]. Therefore, a hybrid approach was adapted here that combines both encoder- and accelerometer-based methods. Since rather small angular vibration amplitudes constitute the transmission error of a helical gear pair, a pair of high precision optical encoders (18,000 pulses/revolution) is employed for measurement of rotational angles θ of each gear. A pair of signal conditioners converts the harmonic encoder signals into TTL square-wave pulse trains with distinct rising and falling edges. Figure 7(a) illustrates the instrumentation that includes the optical encoders, signal conditioners, and a high-speed data analyzer of the encoder-based *DTE* measurement system. The analyzer software uses a pulse timing technique that internally compares the rising edges of each pulse train against a common 100 MHz timer. Combined with the gear pair radii and tooth count input into the analyzer, the timing of the pulses is used to calculate the velocity of each gear. Since the high frequency timer is common to both pulse

Table 1 Design parameters of the helical gear pairs used in the experimental study

Number of teeth	50
Normal module, mm	2.714
Normal pressure angle, degrees	18.224
Pitch diameter, mm	150.0
Base diameter, mm	140.954
Major diameter, mm	variable
Normal circular tooth thickness, mm	4.2
Helix angle, degrees	25.232
Face width, mm	20



(a)



(b)

Fig. 7 (a) Encoder-based and (b) accelerometer-based gear transmission error measurement systems

trains, data of one gear is subtracted from the other to yield the time derivative of *DTE* that is integrated numerically to obtain *DTE* itself. This method allowed measurement of *DTE* with the speed range of 0 to 600 rpm (0 to 500 Hz gear mesh frequency). In addition, an accelerometer based *DTE* measurement system is used within 500 to 4000 rpm (415 to 3300 Hz of mesh frequency). In this method, two diametrically opposed linear accelerometers are mounted near the gear blank at certain radius such that $\ddot{\theta}_i$ and $\ddot{\theta}_j$ can be measured with no gravitational effects. Figure 7(b) shows the gear pairs instrumented with the accelerometers and slip rings. These signals were combined to obtain $d^2(DTE)/dt^2 = r_i\ddot{\theta}_i + r_j\ddot{\theta}_j$ which was integrated twice to obtain *DTE*. Details of this method and the signal processing can be found in [29–31].

With these two measurement systems in place, each of the three gear sets were tested at three different torque values up to 250 Nm. The rotational speed of gear was varied with an increment of nearly 50 rpm and root-mean-square (rms) value of the measured *DTE* was recorded under steady state conditions. At the same time, a dynamic model of the test helical gear-shaft-bearing system was developed as shown in Fig. 8. Here, values of $e_{ij}(t)$ and k_{ij} for each test gear pair was predicted under each load value by using a commercially available gear contact analysis model based on reference [23]. Based on prior experiments using the same machine and test specimens, a modal damping value of 0.03 was considered for the analysis of all three test gear pairs. In Fig. 9, the comparison of measured and predicted rms values of *DTE* is

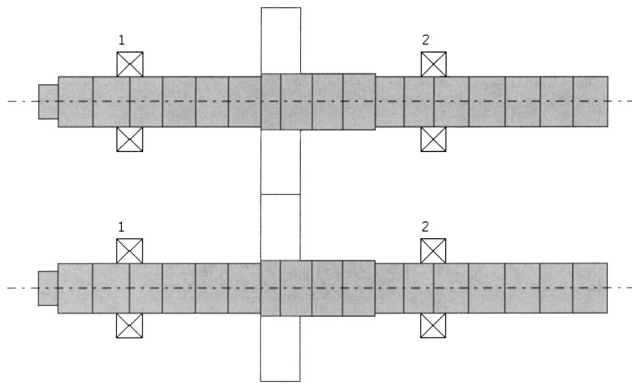


Fig. 8 Dynamic model of the experimental helical gear system

shown for the gear pair with $ICR=1.6$ at 150 and 250 Nm. Similarly, Fig. 10 presents a comparison for the gear pair having $ICR=1.4$ at 150 Nm. As it can be seen from both figures, the measured and predicted values of DTE agree well. The frequencies where the resonances occur are predicted accurately. In addition, the overall shape of the predicted forced response curve and the amplitudes match the experimental data rather well. It is also evident from these figures that the encoder- and accelerometer based measurement methods are in very good agreement in the ranges where data from both methods merge and overlap slightly. It is also clear from the measured data that the forced response is

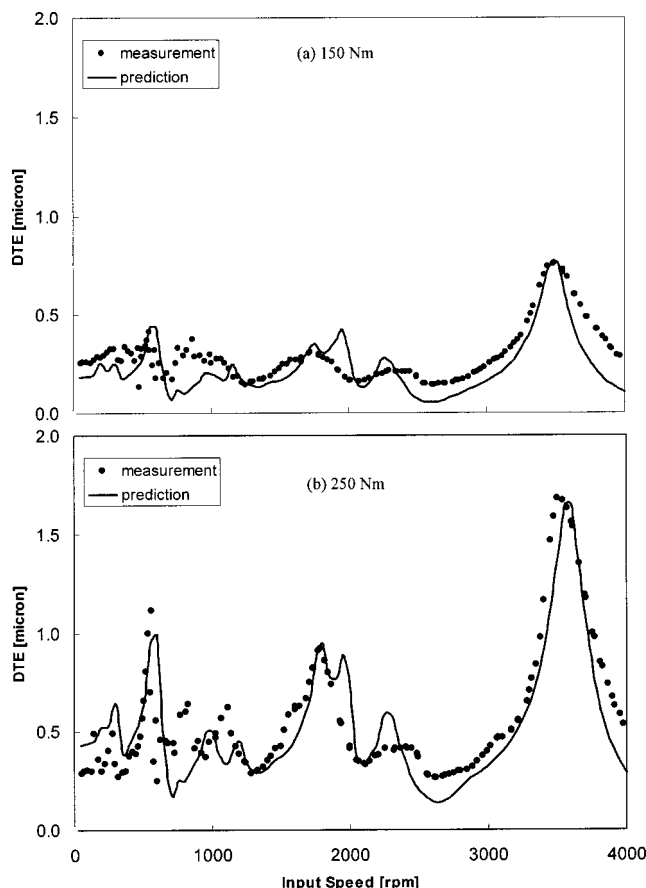


Fig. 9 Comparison of the predicted and measured DTE forced response of the test gear pair with $ICR=1.6$ at (a) 150 Nm and (b) 250 Nm

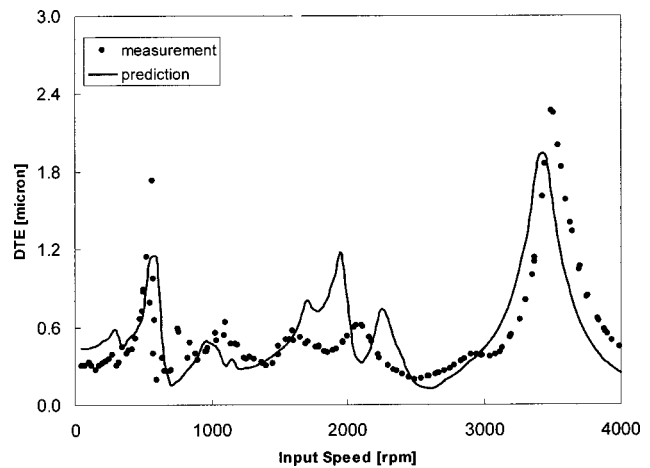


Fig. 10 Comparison of the predicted and measured DTE forced response of the test gear pair with $ICR=1.4$ at 150 Nm

linear. No signs of jump discontinuities of the response curves are observed in measured data, in agreement with the previous published helical gear data [17]. This ensures that the assumption made in neglecting tooth separations is indeed valid for dynamic modeling of helical gears. It is also worthwhile to mention that the use of a loaded static transmission error excitation with a time-averaged mesh stiffness is an acceptable method of modeling helical gear mesh excitations, as it was first suggested by Ozguven and Houser [35]. A similar agreement was achieved for the gear pair having $ICR=1.775$ as well that is not included here due to space limitations.

4 Example Analysis

In the previous section, it was demonstrated that the model agrees well with the experimental data. However, the system considered for the validation was a two-shaft configuration without any complicated shaft details. In this section, a modified version of a three-shaft, 2-gear mesh gear train (see Fig. 11) borrowed from a real-life application will be used to demonstrate influence of some of the relevant design parameters associated with the shafts, gears, and bearings. The nominal values of the system parameters are listed in Table 2. Here the first and second gear pairs have different transverse module values of 1.75 and 2.367,

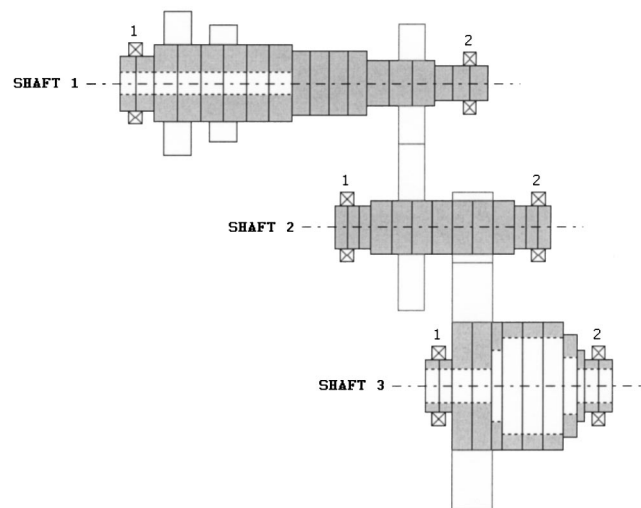


Fig. 11 Dynamic model of the example system

Table 2 Nominal design parameters of the example system ($\gamma_{23}=0$, $\zeta=0.04$)

Shaft Dimensions (mm)									
segment	shaft 1			shaft 2			shaft 3		
	OD	ID	L	OD	ID	L	OD	ID	L
1	41	17	24	32	0	27	40	26	20
2	58	17	102	40	0	106	97	26	30
3	48	0	56	32	0	27	97	53	8
4	34	0	51				97	73	45
5	26	0	39				78	43	10
6							53	26	6
7							40	26	20

Gear Parameters							
Gear	I kg mm ²	J kg mm ²	m kg	pitch dia. mm	Z	face width mm.	hand
1	830	1,660	0.95	91.0	52	18.7	left
2	2,200	4,400	1.22	126.0	72	12.5	right
3	30	60	0.29	52.07	22	17.4	left
4	15,000	30,000	4.67	187.00	79	21.3	right

Bearing Parameters					
Bearing	k_{xx} N/m	k_{yy} N/m	k_{zz} N/m	$k_{\theta_x\theta_y}$ Nm/rad	$k_{\theta_y\theta_z}$ Nm/rad
All	$2(10)^9$	$2(10)^9$	$(10)^9$	$(10)^6$	$(10)^6$

Gear Mesh Parameters				
Mesh ij	k_{ij} (N/m)	$\tilde{e}$ (μ m)	α_{12} (deg.)	
1–2	$3.1(10)^8$	0.5	0	
3–4	$4.14(10)^8$	0.5	0	

respectively. In addition helix angle and transverse pressure angle values are both 20 deg for each gear mesh. Since the list of these parameters is rather long, a complete parametric study of all the parameters involved is not feasible. Therefore, the influence of a small number of parameters will be demonstrated here. The average gear mesh stiffness values and the static transmission error harmonic amplitudes listed in Table 2 are determined by using the same gear contact mechanics model as before [23]. Without loss of generality, only the fundamental harmonics of the transmission error excitations are used. The transmission error amplitudes are rather low since the gears are lightly loaded and hard ground with very little profile errors.

The example system with $\alpha_{12}=0$ and $\alpha_{34}=0$ is considered as the baseline. The model is used to predict the dynamic gear mesh forces at both meshes as $F_{m1}(t)=k_{12}p_{12}(t)$ and $F_{m2}(t)=k_{34}p_{34}(t)$ where $p_{12}(t)$ and $p_{34}(t)$ are relative gear mesh displacements as defined in Eq. (5). The maximum dynamic gear mesh force amplitudes F_{m1} and F_{m2} predicted by the model are shown in Fig. 12. The resonance peaks at 900, 1,750 and 3,000 rpm are evident from this figure. These resonance frequencies occur when the gear mesh frequencies ω_{m1} and ω_{m2} are equal to a number of natural frequencies including f_2 , f_5 , f_8 and f_{26} . All of these modes exhibit coupled transverse-axial-rotational motions with non-zero relative gear mesh displacements.

As the first system parameter, the length of the second shaft L_2 that carries gears 2 and 3 is varied incrementally from 120 mm to 250 mm. This change is achieved by adding length to each segment of the second shaft (between the gears and between a gear and a bearing) proportionally. In Fig. 13, F_{m2} is plotted against the input speed for different L_2 values. Here, the influence of L_2 is rather drastic. The peaks at 1,000 and 3,000 rpm are shifted to the left and the peak amplitudes are reduced significantly with increasing L_2 . The same effect is observed on F_{m1} as well.

Influence of shaft angles on the dynamic response is illustrated in Fig. 14. Here, α_{34} is varied from -49.09 to 98.18 deg incre-

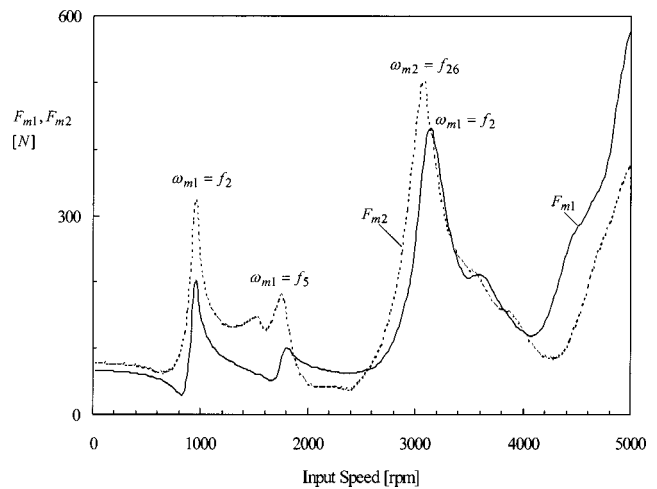


Fig. 12 Dynamic gear mesh force amplitudes F_{m1} and F_{m2} of the example system

mentally while $\alpha_{12}=0$. The particular α_{34} values chosen represent integer multiples of the base pitch of gear 3 such that the phase angle between $e_{12}(t)$ and $e_{34}(t)$ is not influenced by α_{34} according to Eq. (10). In Fig. 14, there is a significant amount of change in the resonance peak amplitude of F_{bx} of the second bearing on the second shaft while the change in the natural frequencies is negligible. The value of F_{bx} reaches its maximum at $\alpha_{34}=98.18$ deg and its minimum at $\alpha_{34}=0$.

As a representative of bearing parameters, stiffness coefficients of every bearing of the system in bending direction $k_{\theta_y\theta_y}=k_{\theta_x\theta_x}$ are varied equally. Figure 15 illustrates the change in F_{m2} as a function of $k_{\theta_y\theta_y}=k_{\theta_x\theta_x}$. Here, the resonance frequencies are shifted to the right when $k_{\theta_y\theta_y}=k_{\theta_x\theta_x}$ is increased, indicating that the corresponding natural frequencies are also increased. The resonance peak amplitudes of F_{m2} are also increased with the same change. While the bearing force amplitudes in radial and axial directions were not very sensitive to the value of $k_{\theta_y\theta_y}=k_{\theta_x\theta_x}$, bearing bending moments $M_{by}=k_{\theta_y\theta_y}\theta_{by}$ and $M_{bx}=k_{\theta_x\theta_x}\theta_{bx}$ were influenced drastically.

Finally, the influence of the hand of the helical gears on dynamic bearing forces are predicted. All possible combinations of the gear hand are considered: LR–LR, LR–RL, RL–RL and RL–LR. For instance, case LR–LR represents a system with

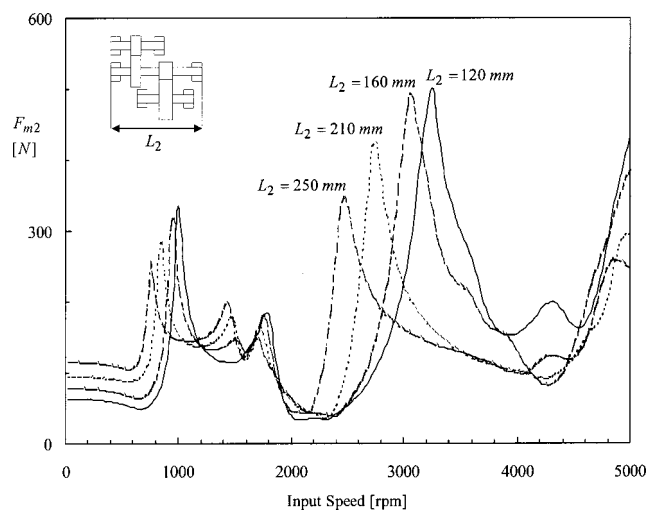


Fig. 13 Influence of the length of the second shaft on the dynamic gear mesh force amplitudes F_{m2}

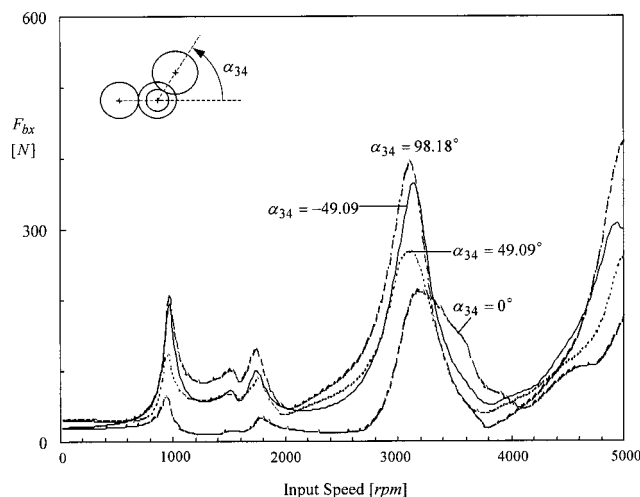


Fig. 14 Influence of shaft angle α_{34} on F_{bx} of the second bearing of the second shaft

left-hand first and third gears and right-hand second and fourth gears. In Fig. 16, the influence of the gear hand on the resonance peak amplitudes of F_{by} (of the same bearing as Fig. 14) is quite significant. The same is true for F_{bx} as well, while F_{bz} is affected very slightly by the hand of gears.

In summary, above limited set of parametric studies indicate that the dynamic behavior of multimesh, multi-shaft gear-shaft-bearing systems is rather complex. A large number of parameters influence the dynamic behavior in different levels and ways. Therefore, generalized guidelines are not readily available based on an analysis of a single system.

5 Conclusion

A dynamic model of a multi-shaft helical gear reduction unit was proposed in this study. The model consists of a finite element model of shaft structures combined with a three-dimensional discrete model of helical gear pairs. Bearing and housing flexibilities were included in the model as well. Results of an experimental study were used to validate the model predictions. The experimental data also revealed that the key assumption regarding the linearity of dynamic behavior and the use of a time-invariant gear mesh flexibility are also valid for modeling of helical gears while

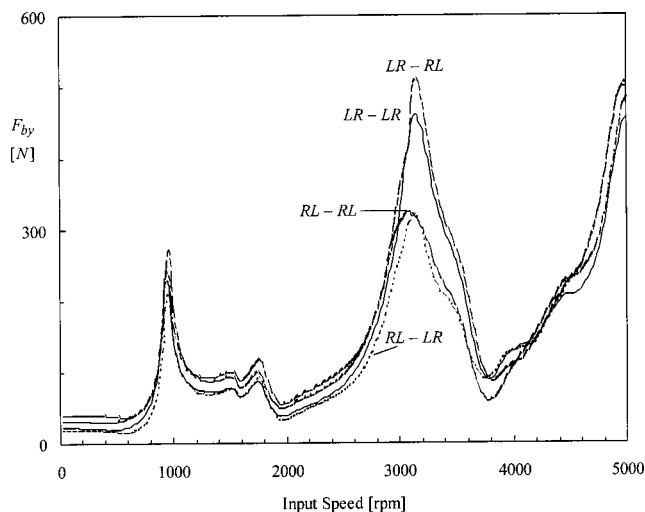


Fig. 16 Influence of gear hand configurations on F_{by} of the second bearing of the second shaft

this might not be true for spur gears. Free and forced vibration characteristics of an example system were studied to demonstrate the influence of a number of system parameters including shaft angles, shaft dimensions, bearing stiffness, and the hand of the gears.

The model has the capability of including most of the design parameters relevant to bearings, shafts and gears as described. In this study, it was shown that a large amount of design parameters can influence the overall dynamic behavior of the system. This model can help the designers not only to predict the dynamic behavior of the system but also to come up with the most favorable configuration for the most desirable dynamic behavior, especially when used in early stages of cross-section design. The model can also be used for trouble shooting for cases when an existing product exhibits undesirable dynamic behavior. However, limited parametric studies suggest it was also demonstrated that generalized design guidelines may not be readily available for such systems as a large number of parameters interact with each other to influence the dynamic behavior.

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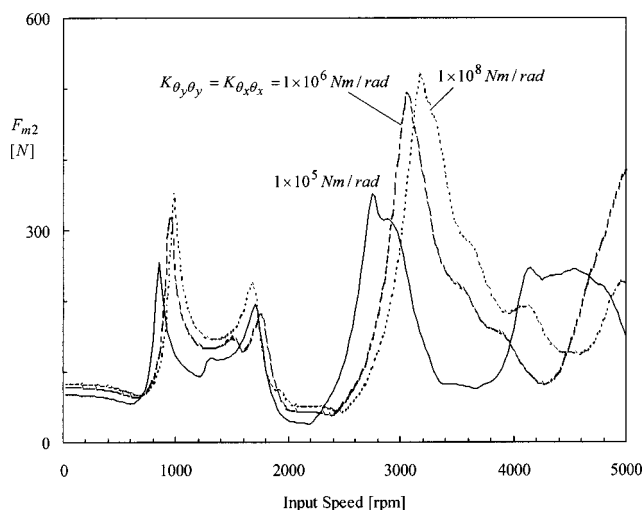


Fig. 15 Influence of bearing stiffness coefficients $K_{\theta_y \theta_y} = K_{\theta_x \theta_x}$ on F_{m2}

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